A four-inequality limit for the Part II geometry of the 7/8 zero-free region

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Abstract

Part II of the September 30, 2026 quasi-Riemann paper (OAI) proves a zero-free region $\operatorname{Re} s > 7/8$ for one fixed geometry $(b, l_x, \ell) = (1/8, 17/48, 1/6)$. We ask how far the same exponent inequalities could go for any geometry. Four of them, combined with exact rational multipliers, give $s_0 \geq 4638705/5301659 = 0.8749535\ldots$, so the architecture has less than $4.7 \cdot 10^{-5}$ of room below 7/8. Assume the mean-square hypotheses MS(κ) for all $\kappa \in [\kappa_0, 1]$, with the uniformity of OAI's (17.87). Under them the same method gives exact lower bounds that match the numerical optima to about 10^{-5}, and a κ_0-free bound $s_0 \geq 167/192 = 7/8 - 1/192$. Both bounds survive dropping the condition $M + \ell \leq 1$ once its row-norm cost is accounted for. A smaller floor bin gains at most 1/320. The statements concern this exponent architecture only. They are not barriers for other methods.

1 The inequalities

Line numbers refer to `paper.tex` of OAI. Write $l_y = l_x + b$, $M = l_x + l_y$, $h = 1 - l_x + \ell$, $z_0 = 17/50$, and let s_0 be the target boundary (σ_0, β_* in OAI). An independent reader derived each inequality below from the source before comparing it with our transcription. Each one matches the paper or is weaker than it, so it is a necessary condition for any geometry for which the argument closes.

(N) $b, l_x, \ell \geq 0$ and $l_x \geq \ell$. The paper requires $P_a \geq 1$ (line 8344) and $l_x - d \geq 3/16$ for $d \in [0, \ell]$ (line 8594).

(S) $M + \ell \leq 1$. The paper fixes $M + \ell = 1$ (line 6873).

(L) Low estimate (Prop. probe-low, lines 8564–8642). The bound $l_x/2 + b/12 + \max_{d \in [0,\ell]}(-d + \operatorname{loss}(d)/2)$ must not exceed $C(s_0) = l_x/2 + s_0 - 1 + h/6$. Taking $d = 0$ and dropping $\operatorname{loss}(0) \geq 0$ gives

$$s_0 \;\geq\; \tfrac{5}{6} + \tfrac{b}{12} + \tfrac{l_x - \ell}{6}.$$

(H) High exponent (eq. stage-high-exponent, line 6092). At row exponent $d = h$, which is the worst case on $[1/2, h]$ (line 16133), with frequency bin δ and $g = x\delta\ell$:

$$a - s_0 + h(z_0 - \tfrac{1}{6}) - a\, l_y - \tfrac{\ell}{2} + x\delta\ell + h\big(R + \tfrac{\delta}{2} - z_0\big) \leq 0, \qquad a = \tfrac{1+\delta}{2},$$

for every $\delta \in [1/50, \kappa]$, $\kappa = 2s_0 - 1$, and $x \in [0, 1/2]$. Here $R = R^*(\delta, x, \ell/h)$ is the row-count exponent of Prop. 19.2 with the long-range cost of Lemma 17.6, an infimum over the cutoff t of a maximum over the length r. We use (H) only at $x = 1/2$. Lemma 1 evaluates R^* for that case, and it does not cover other x.

(F) Floor bin $\delta_0 = 1/50$ with $R = 1$ (eq. floor-bound, line 15845): (H) with $\delta = 1/50$, $x = 1/2$, $R = 1$.

In (H) and (F) the terms in z_0 cancel, so the bounds below do not depend on the choice $z_0 = 17/50$.

Relaxation below 7/8**.** Several lemmas of OAI are stated for $\sigma_0 \geq 7/8$. We do not use any of their conclusions. We only use inequalities (L), (H), (F), each of which says that an exponent the argument must make negative is negative, together with the count of Prop. 19.2 at capacity $c = 1/(6\max(3/4, \kappa)) = 2/9$, the largest capacity Lemma plain allows. A hypothetical extension of the proof to $s_0 < 7/8$ needs at least these conditions, and granting it the unproved lemmas only enlarges its feasible set. So the infeasibility below is a valid limit for the architecture.

2 The count lemma

Lemma 1. *Let $2/3 \leq \kappa_0 \leq 1$, $0 < \delta < 5\kappa_0/6$, $x = 1/2$, $q = \delta/2$, $c = 2/9$. For $t \in [1, 3/2]$ and $r \in [t - 1/2, t]$, $m = t - r$, put $f = e(r) - \delta r$ for $r \geq 1$, with $e(r) = 1$ for $r \leq 1/\kappa_0$ and $(1 + 5\kappa_0 r)/6$ otherwise, and $f = \min\big(1 - \delta r - 2qz_i,\ 1 - 2\delta m - 2qz_p\big)$ for $r < 1$, with $z_i = \min((1 - r)/2, \sigma)$, $z_p = \min(c(1 - 2m), \sigma)$, σ the supply cap. Then $R^*(\delta, \sigma) = \inf_t \sup_r f \ \geq\ R_\infty(\delta)$, where*

$$R_\infty(\delta) = 1 - \frac{\delta(16 + 14t^*)}{37}, \qquad t^* = \frac{5/6 - 16\delta/37}{5\kappa_0/6 - 23\delta/37},$$

if $1/\kappa_0 \leq t^* \leq 3/2$, and $R_\infty(\delta) = 1 - \delta$ otherwise.

Proof. Lowering the caps raises f, so $R^*(\delta, \sigma) \geq R^*(\delta, \infty)$. Without caps, $z_i = (1 - r)/2$ and $z_p = \frac{2}{9}(1 - 2m)$. For $r < 1$ the two entries of the minimum are $1 - \delta/2 - \delta r/2$, decreasing in r, and $1 - 2\delta/9 - 14\delta m/9$, increasing in r. They meet at $r = (28t - 5)/37 \in [t - 1/2, 1)$, so $\sup_{r<1} f = S(t) = 1 - \delta(16 + 14t)/37$, a decreasing function with $S(3/2) = 1 - \delta$. For $r \geq 1$, $\sup f = \max\big(1-\delta,\ L(t)\big)$, where $L(t) = 1/6+t(5\kappa_0/6-\delta)$ is increasing and present only for $t > 1/\kappa_0$. Since $L(1/\kappa_0) = 1 - \delta/\kappa_0 < S(1/\kappa_0)$, the infimum over t of $\max(1 - \delta, S, L)$ is $S(t^*) = L(t^*)$ when $t^* \leq 3/2$, and $1 - \delta$ otherwise. □

At $\kappa_0 = 1$ this gives $R_\infty(19/50) = 55889/82850$ and $R_\infty(2/5) = 2129/3245$. The values agree with the reduced-model code to within its grid error ($6 \cdot 10^{-5}$).

3 The bound at $\kappa_0 = 1$

Theorem 2. *Let $\kappa_0 = 1$. Every geometry (b, l_x, ℓ) that satisfies (N), (S), (L) and (H) has*

$$s_0 \ \geq\ \frac{4638705}{5301659} = 0.8749535\ldots.$$

Proof. If $s_0 < 5/6$, (L) and (N) give $s_0 \geq 5/6$, a contradiction. Let $s_0 \geq 5/6$. Then $\kappa \geq 2/3$, so $\delta = 19/50$ and $\delta = 2/5$ are admissible in (H). With Lemma 1 and $R \geq R_\infty$, (H) becomes

$$\begin{aligned} &(\mathrm{H}_1) \quad s_0 \geq \tfrac{172483}{124275}(1 - l_x) + \tfrac{48208}{124275}\ell - \tfrac{69}{100}b, \\ &(\mathrm{H}_2) \quad s_0 \geq \tfrac{13526}{9735}(1 - l_x) + \tfrac{3791}{9735}\ell - \tfrac{7}{10}b. \end{aligned}$$

Write each inequality as $c_b b + c_l l_x + c_e \ell + c_s s_0 \leq \rho$ and add them with the multipliers in Table 1. The coefficients of b, l_x and ℓ cancel exactly and the coefficient of s_0 is -1, which gives the bound. □

Inequality	multiplier λ	c_b	c_l	c_e	c_s	ρ
(L)	4307228/5301659	1/12	1/6	−1/6	−1	−5/6
(S)	331477/5301659	1	2	1	0	1
(H_1)	1706710/15904977	−69/100	−172483/124275	48208/124275	−1	−172483/124275
(H_2)	116053/1445907	−7/10	−13526/9735	3791/9735	−1	−13526/9735
$\sum \lambda \cdot$		0	0	0	−1	−4638705/5301659

Table 1: Exact multipliers for Theorem 2 (`symproofk.py` checks every entry).

The obstruction is a three-way balance. (L) rewards large l_x and small ℓ, (H) charges for small $1 - l_x$ through the count R, and (S) ties the three lengths together. The floor, the intermediate range and the length condition $l_y - d - 11b/6 \geq 1/12$ do not enter. The audit marked the last two as stronger than the paper needs, so it matters that the proof does not use them.

4 Conditional variants

Assume MS(κ) for every $\kappa \in [\kappa_0, 1]$, with the uniformity over twists, symbol signs, rowwise profiles and heights required in OAI's (17.87). A single estimate at κ_0 is not enough. Then $e(r) = 1$ for $1 \leq r \leq 1/\kappa_0$, and the copy argument of Lemma 17.6, run from MS(κ_0), gives $e(r) \leq (1 + 5\kappa_0 r)/6$ beyond. This changes only the long-range cost in Lemma 1. The same method gives the following exact bounds. Each certificate has four rows, chosen from (L), (S), (H) at $x = 1/2$, (F) and $b \geq 0$. Appendix A lists their multipliers.

κ_0	rows used	exact bound	decimal	numerical optimum
1	L, S, H(19/50), H(2/5)	4638705/5301659	0.8749535	0.87497
97/100	L, S, H(19/50), H(2/5)	49696303/56869303	0.8738687	0.87386
19/20	L, S, H(19/50), H(9/25)	7753639/8880447	0.8731136	0.87312
9/10	L, S, H(17/50), $b \geq 0$	336893/386718	0.8711593	0.87117
4/5	L, S, F, $b \geq 0$	167/192	0.8697917	0.8698

Table 2: Lower bounds under MS(κ) for all $\kappa \in [\kappa_0, 1]$. The numerical optima are from the reduced model, without the intermediate convenience bound $76/75 - 2\delta/3$.

Theorem 3. *For every κ_0, the inequalities (N), (S), (L) and (F) force $s_0 \geq 167/192 = 7/8 - 1/192$.*

Proof. (F) reads $s_0 \geq \frac{203}{150}(1 - l_x) + \frac{53}{150}\ell - \frac{51}{100}b$. Adding (L), (S), (F) and $-b \leq 0$ with multipliers 103/128, 25/384, 25/128 and 25/768 cancels b, l_x and ℓ and gives $s_0 \geq 167/192$. None of these rows involves the count, so the bound holds for any mean-square input. □

Remark. No mean-square improvement can move this architecture below $7/8 - 1/192$. Going further would need a change in the floor bin $\delta_0 = 1/50$, which is tied to the Euler region $x_r \geq 51/100$, in the low estimate, or in the relation $M + \ell = 1$.

5 Robustness

The condition $M + \ell \leq 1$ is not needed. OAI uses $M + \ell = 1$ once, in the row-norm lemma (line 8246, "Since $M' + \ell' - 1 = -3d$"). Put $\sigma = M + \ell - 1$. Without the condition the available dual length

satisfies $T_d \le H-3d+\sigma$. Retained dual dyads have $v+\ell_b \le y \le T_d$, so $\max(H, v+\ell_b) \le H+(\sigma-3d)_+$, and the row-norm exponent grows by at most $(\sigma-3d)_+$. The loss formula $M' + (1+3\ell'-2M')/4$ comes from the definition of T_d and does not involve σ. After Cauchy–Schwarz, the low estimate at $d=0$ becomes

$$(\mathrm{L}_\sigma) \qquad s_0 \;\ge\; \tfrac{5}{6} + \tfrac{b}{12} + \tfrac{l_x-\ell}{6} + \tfrac{1}{2}(b+2l_x+\ell-1)_+.$$

Replacing (S) by the linear branch of (L_σ) gives the same bounds. At $\kappa_0 = 1$ the multipliers of (L), (L_σ), (H_1), (H_2) are 3644274/5301659, 662954/5301659, 1706710/15904977, 116053/1445907, with bound 4638705/5301659. For the floor bound they are 259/384, 25/192, 25/128, 25/768, with bound 167/192. A geometry with $M+\ell > 1$ would help only if the penalty slope 1/2 fell below about 0.077.

The floor bin. With a general floor bin $\delta_0 = 2a-1$, the rows (L), (S), (F) and $b \ge 0$ give exactly

$$s_0 \;\ge\; \frac{2}{3} + \frac{1+2\delta_0}{5+6\delta_0},$$

which is 167/192 at $\delta_0 = 1/50$ and tends to 13/15. Lowering a below 51/100 costs nothing in the local correction (lines 4144–4170). The good-prime exponent $4-6x_r-6z_r$ is -1.07 at $x_r = 0.505$, $z_r = 17/50$, and the ramified exponents are at most $1-2x_r-\epsilon_0$. In the full system, though, the floor binds only when $\kappa_0 \lesssim 0.8$. Numerically, the lower bound at $\kappa_0 = 1$ and 0.9 is unchanged for $\delta_0 = 0.01$ and 0.004, and at $\kappa_0 = 0.8$, $\delta_0 = 0.004$ the count binds again at 0.86813.

A bilinear low estimate. If the low bound $l_x/2 + b/12$ improved to $l_x/2 + b/12 - \eta l_x$, the bound would fall steadily. About $\eta = 0.07$, a saving of roughly $Z^{-0.03}$ over Cauchy–Schwarz, would reach 0.85, and no other constraint binds before $\eta = 0.2$. However, the low-separated sum $Q^{-1/2}\sum_m w_m A_m B_m$ equals the probe I_η itself (eq. low-separated, line 3470). An unconditional saving over Cauchy–Schwarz would itself be the stronger low estimate, so it cannot be supplied as a separate input. Measured cancellation at finite scales cannot establish the uniform saving either. It reflects the zeros present at those scales: without zeros off the line the ratio is about $Z^{C(1/2)-l_x/2-b/12} \approx Z^{-0.36}$. This does not rule out a new unconditional bilinear argument, but such an argument would amount to a new way of dividing the probe between the low and high ranges.

6 Other approaches tested

We also tested several modifications of the architecture. Each item below records the specific obstruction we met and says whether it is an exact argument or a numerical finding. These are limits of the approaches tested, not impossibility results for other methods.

- *Mean-square input, floor bin, $M+\ell \le 1$* (exact; Sections 4 and 5). Mean-square input alone stops at 167/192. A smaller floor bin gains at most 1/320, and only for $\kappa_0 \lesssim 0.8$. Relaxing $M+\ell \le 1$ is paid back by the row norm.

- *Cancellation across floor rows* (analytic). After Poisson summation, the common-contour version leads to an off-diagonal problem at row range $H \asymp D$, where the available generic estimates give no saving. This does not exclude estimates that use more arithmetic structure.

- *Decomposition by top row dyads* (analytic). The filter to sixth-power-free rows costs $Z^{h/6}$, which is what the decomposition gains.

- *Free ceiling on the Poisson frequencies, and savings in the floor-row count* (numerical, reduced model, not certified). Neither gains anything at $\kappa_0 = 1$ (0.87498 against 0.87497), and the gain is below 0.002 at $\kappa_0 = 0.8$.

- *A cubic probe* (exact at the exponent level). The cubic pole at $z = 1/3$ fits the signal budget, and the generic cubic Gram cost ($P_a^{1/3}$ in place of $P_a^{1/6}$) is small. However, the generic cubic large sieve controls the fourth moment of Lemma plain only up to length $M/2$, short of the length M the argument needs, with a deficit of $m/3$ coming from the cube copies. Moreover, assume the uncentred threshold is $(1 - 1/k)M$ and write s for the common-support saving ($s = 2/3$ for $k = 6$). With one comparison length for every product length A, as in OAI ($L = M/4$), the two branches of the induction need $M/(k \min(s, 1)) \le L < (\frac{1}{2} - \frac{1}{k})M$, which is possible only for $k > 4$. If L may depend on A, the condition at the endpoint $A = M$ becomes $\min(s, 1) > 2/(k - 1)$. That still excludes $k = 3$ for every $s \le 1$, so a cubic probe needs a different fourth-moment argument, not a transplant of the sextic one. The relaxed condition does not exclude $k = 4$ (with $s > 2/3$) or $k = 5$ (already with $s = 2/3$). For those orders, the input that is missing is the explicit theta coefficients the reflection identity uses. They are known for orders 2 and 3, the factors of 6, but not for 4 or 5. This ledger is exponent-level only.

- *A bilinear low estimate* (Section 5). The sum is the probe itself, so a saving would have to be an unconditional estimate, which amounts to a new way of dividing the probe.

Within the audited exponent model, none of the modifications tested moves the boundary materially. Going beyond this architecture appears to need a new signal mechanism or a different low-separation estimate.

Reproducibility. The script `symproofk.py` checks every statement in exact rational arithmetic. The script `certs_cond.py` re-verifies all seven certificates. It writes them, with every row coefficient, to the machine-readable file `cond_certificates.json`. The checker `certcheck.py` independently verifies the $\kappa_0 = 1$ result for $s_0 \le 437/500$ with a 1013-box certificate.

A Exact multipliers of the remaining certificates

Each block lists the four rows, their multipliers, and for (H) the value $R_\infty(\delta)$ from Lemma 1 at that κ_0. With $R = R_\infty$, the row coefficients follow from the formulas in Section 1. In every block the coefficients of b, l_x and ℓ cancel exactly and that of s_0 is -1. Table 1 gives the case $\kappa_0 = 1$.

row	multiplier λ	$R_\infty(\delta)$
$\kappa_0 = 97/100$: $s_0 \ge 49696303/56869303$		
(L)	46109803/56869303	
(S)	3586500/56869303	
(H) $\delta = 19/50$	30799925/170607909	211961/317525
(H) $\delta = 2/5$	1478575/170607909	1151/1775
$\kappa_0 = 19/20$: $s_0 \ge 7753639/8880447$		
(L)	2396745/2960149	
(S)	563404/8880447	
(H) $\delta = 19/50$	1324609/8880447	10749/16225
(H) $\delta = 9/25$	365603/8880447	30697/45025
$\kappa_0 = 9/10$: $s_0 \ge 336893/386718$		

(L)	207987/257812	
(S)	49825/773436	
(H) $\delta = 17/50$	49825/257812	102997/149475
$b \geq 0$	3347/1546872	
$\kappa_0 = 4/5$: $s_0 \geq 167/192$		
(L)	103/128	
(S)	25/384	
(F)	25/128	
$b \geq 0$	25/768	
$\kappa_0 = 1$, without (S): $s_0 \geq 4638705/5301659$		
(L)	3644274/5301659	
(L_σ)	662954/5301659	
(H) $\delta = 19/50$	1706710/15904977	55889/82850
(H) $\delta = 2/5$	116053/1445907	2129/3245
$\kappa_0 = 4/5$, without (S): $s_0 \geq 167/192$		
(L)	259/384	
(L_σ)	25/192	
(F)	25/128	
$b \geq 0$	25/768	